

Substituting Reference σ -Profiles Degrades Solubility Prediction and Separates COSMO-SAC Error from Input Error

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Abstract

Physics-informed models route predictions through interpretable physical quantities, expecting that accurate reference values for them improve accuracy. We measure that for solid-liquid solubility. A graph neural network predicts a molecule’s charge-density profile, the fixed thermodynamic model COSMO-SAC turns a pair of profiles into an activity coefficient and hence a solubility, and a quantum-chemical database tabulates the same profiles. Grounding names two opposite operations on it, with opposite signs: supervising the predicted profile against it during training improves solubility accuracy over three seeds by 0.20 in mean absolute error; substituting the reference for the learned profile at inference degrades it by 0.41. It is also an instrument: on infinite-dilution activity data it bounds the model’s own error from below and its inputs’ from above. The bound on the inputs barely moves between chemical strata while the total error differs across them by more than an order of magnitude, so the comparison is a map over 59 strata, not one verdict. Under one admissibility rule applied alike to every stratum, fifteen are large enough to bound and six keep their margin’s sign under every publication deletion; those six are three row sets, and on one—the glycol-ether solvents, 182 measurements over 43

molecule pairs—the model’s error is the larger term, by +2.04 (90% interval [+1.27, +2.83]). The map reports the rest without establishing it. Trained through the misspecified model, the learned profile departs from its reference along a shared direction. Grounded latents are not reliably interpretable when the model consuming them is misspecified.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug can be formulated as a tablet or an injection, which solvent a crystallisation is run in, and how a product is purified. It has to be measured one solute, one solvent and one temperature at a time, so most of the combinations that matter industrially have never been determined, and independent determinations of the same system disagree by enough to put a floor under what any predictor can achieve. Dissolving a solid also trades the cost of breaking up its crystal against the cost of mixing the freed molecules into the solvent, so a predictor has to be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph; FastSolv¹ does

so with a deep neural network and leads on extrapolation to new solutes. They are the more accurate there and they are opaque: the prediction arrives as a single number, with no account of which part of the chemistry it captured. Physics-informed models instead predict the thermodynamic quantities the physical picture names—an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal—and combine them through a fixed equation, as SolProp² does with a thermodynamic cycle of separately learned components. What that buys is decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data a direct regressor has no way to use. On accuracy the black box leads and the physics-informed cycle trails it.

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. For one class of intermediate that can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. It needs testing, because when a predictor of this shape is inaccurate either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies: a more faithful model in the first case, better representations or more data in the second.

Learned surrogates for the activity coefficient have advanced quickly. Winter et al.³ predict infinite-dilution activity coefficients $\ln \gamma_2^\infty$ directly from SMILES with a language model, and Sanchez-Medina et al.⁴ embed a Gibbs–Helmholtz constraint in a graph network for their temperature dependence; both make the activity coefficient the regression target rather than retaining a mechanistic model that consumes a predicted molecular property. A second line keeps the mechanistic model and learns its input. COSMO-SAC⁵ is a segment-activity reformulation of Klamt’s COSMO-RS, and the σ -profile it consumes—screening-charge density over a molecule’s cavity—is that framework’s construct, tabulated from quantum chemistry.⁶

Predicting it from structure removes that calculation from the pipeline; TeNNet-SAC⁷ does so while also learning the segment-activity kernel, and gains from truer profiles.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning,⁸ whose dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,⁹ gray-box hybrids composing learned components with fixed mechanistic maps,¹⁰ and operator learning with a trainable correction to a misspecified equation.¹¹ The prevailing expectation is that more physics improves generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse.¹² In neural PDE solvers, Mohan et al.¹³ show that a learnable term added to a fixed discretised operator absorbs truncation artifacts rather than physics, so the apparent interpretability is illusory—though their error source is numerical and has no external reference. The same concern appears in concept-bottleneck models as *concept leakage*, a learned concept silently encoding information beyond its nominal meaning.^{14–16} Recent work traces one route to leakage to the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence the head cannot represent,¹⁷ and overwriting a frozen head’s leaked concepts with ground truth can then degrade accuracy.¹⁸ The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O’Hagan¹⁹ and its calibration critique.²⁰ Four further literatures bear on it—forecast decomposition, quasi-maximum likelihood, identifiability, and the validity of composed pipelines—and are reviewed in §S12.

What this literature lacks is a way to *measure*, rather than diagnose, whether a fixed physical model or its learned inputs limits a composed predictor’s accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after; in the concept-bottleneck case no external reference exists against which either question could be settled. This work supplies one—a reference σ -profile for solubil-

ity, tabulated substituent constants for a second chemical domain—and holds the mechanistic kernel fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them, which leaves the network’s own intermediate with a tabulated counterpart to be held against. That counterpart can enter the pipeline at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). The thermodynamic model is implemented in differentiable form so that both routes are open on a single trained network, and the tabulated values are put to a third use, dividing the fixed model’s error between the map and the inputs handed to it.

Only one of the two routes helps, and the reason is a property of the fixed map. Solubility data cannot say which component is at fault, because one measurement constrains the pair jointly; activity coefficients can, because both molecules carry a tabulated distribution and the two error terms separate. Where they separate, an ordering decides the remedy: a map whose own error stands above what its inputs fail to carry does not become more accurate when handed a better input, but more exposed, the accurate input no longer masking the bias the map introduces. That ordering is a property of the chemistry the map is asked about, so the instrument returns a map over chemical strata rather than a verdict, and one rule applied alike to every stratum decides what the map establishes. Two consequences follow. A learned intermediate trained through a misspecified map cannot be read as the physical quantity it is named after: the predicted distribution drifts away from the tabulated one, along a direction shared across molecules. And substituting a reference value for it at prediction time is not a safe improvement but a change of regime, which is why the substitution here costs more than the supervision gained. Neither consequence is peculiar to charge distributions: tabulated sub-

stituent constants behave the same way inside a fixed Hammett relation for acid dissociation, improving the aromatic series that relation was built for and degrading the ones outside it.

2 Computational methods

2.1 The composed predictor and its error split

For a solute s in solvent v at temperature T , the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades two costs against each other—breaking the crystal lattice (Φ) and mixing the freed molecules against non-ideal solute-solvent interactions ($\ln \gamma_2$)—and solid-liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = - \underbrace{\frac{\Delta H_{\text{fus}}}{R} \left(\frac{1}{T} - \frac{1}{T_m} \right)}_{\Phi(T) \text{ (crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with the constant- ΔC_p (heat-capacity-of-fusion) correction omitted for clarity. The activity term is produced by a *differentiable closure* g : a graph encoder predicts a σ -profile—the distribution of screening (surface) charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC,⁵ a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln \gamma_2$. Inside that map the segment-interaction energy—the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other—is the closure’s *kernel*, the part the published revisions rewrite, so a closure variant is named for the year of its kernel: the deployed *2002 kernel*, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write z^* for the *reference* physical latent, the σ -profile tabulated in the VT-2005 database,⁶ and $g(z^*)$ for the closure output when fed it; the model’s own head instead predicts a *learned* profile $\hat{z}$ and evaluates $g(\hat{z})$. Feeding z^* in place of $\hat{z}$ (a σ -oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^*)$ should

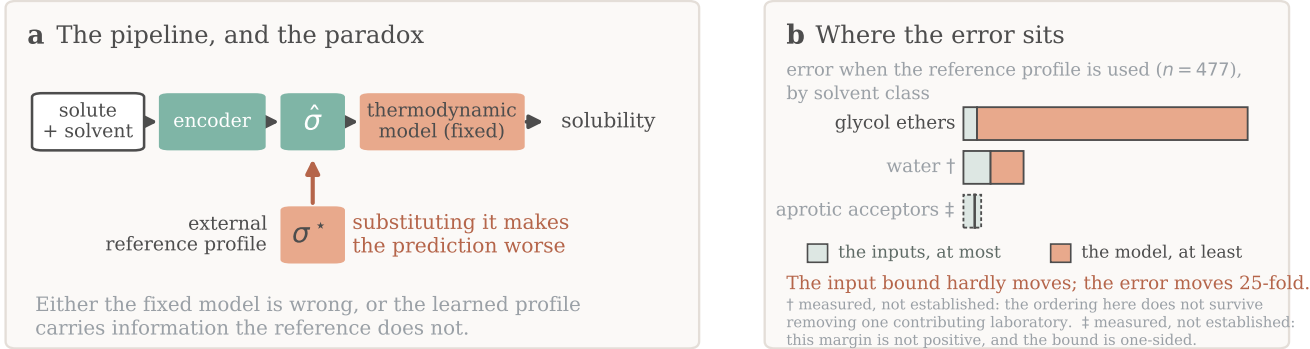


Figure 1: **Overview.** (a) A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility; the arrow marks the evaluation-time substitution of the reference σ^* , which makes it worse. (b) The counterpart error on infinite-dilution activity data, by solvent class; only the unmarked class is established, each mark carrying its own reason, and each block is a one-sided bound rather than an interval. Training-time supervision is not drawn.

be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*; a *DirectGNN* control arm drops the closure and emits $\ln x_2$ directly (Fig. S1), and the two were tuned separately, so the gap between them is not attributable to the bottleneck alone (§2.4; per-arm settings in §S1).

Everything below is stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical intermediate $z = h(x)$, against a matched free predictor that shares the encoder and drops g ; solubility is one instance, §S7 a second (pKa through a fixed Hammett closure) and §S11 a controlled synthetic family. Nothing in Lemma 1 or in §S2 is specific to COSMO-SAC.

The crystal/activity split of Eq. (1) is not identifiable from solubility alone, which is why an unconstrained head can silently absorb crystal error into the activity term—an indeterminate component end-to-end training leaves free rather than projecting out, closed only by a rank-completing external constraint (Lemma S1, §S2). That is enabling background: it explains why an under-determined factor needs external single-component data and why compensation between the factors is expected, and says nothing by itself about whether grounding helps.

2.1.1 The closure–insufficiency decomposition

Let m be the activity target—the experimental infinite-dilution activity coefficient $\ln \gamma_2^\infty$ —and $g(z^*)$ the closure evaluated on the true latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds). *For a fixed (non-fitted) closure g , with z^* denoting the closure’s entire argument (the profile pair together with the temperature wherever the closure reads one),*

$$\underbrace{\mathbb{E}[(m - g(z^*))^2]}_{\text{ref.-input MSE}} = \underbrace{\mathbb{E}[\text{Var}(m \mid z^*)]}_{B_{\text{insuff}} \text{ (irreducible)}} + \underbrace{\mathbb{E}[(\mathbb{E}[m \mid z^*] - g(z^*))^2]}_{B_{\text{closure}} \text{ (decoder bias)}}. \quad (2)$$

The cross term vanishes identically, which is why z^ must be the closure’s whole argument and not a part of it (proof, §S2). The split is exact with no fitting of the closure g , so the total and B_{closure} ’s Jensen lower bound are fit-free. $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^*)]$, however, must be estimated with a conditional-variance smoother whose choice affects the verdict (§2.5).*

B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O’Hagan,^{19,20} and its dominance over B_{insuff} is

the empirical content of the claim that the accuracy ceiling is set by the closure rather than by the inputs. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and z^* externally observed, so the map’s error is *attributed* rather than corrected—available only where an external latent oracle exists, not in gray-box hybrids (§S12). **Exactness is contingent on g being fixed:** for a *fitted* decoder the cross term does not vanish and a plug-in point split is invalid, so we report *one-sided bounds* (proved in §S2):

- $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$ by Jensen: a constant offset between the decoder’s mean output and the target is misspecification, not input insufficiency, and this bound needs *no smoothness assumption*. Its one requirement is that m and g share the $\ln \gamma_2^\infty$ reference state, which the closure validation of §2.3 establishes.
- $B_{\text{insuff}} \leq \mathbb{E}[\text{Var}(m \mid \text{bin}(g(z^*)))]$, a valid *upper* bound (law of total variance, since a binning of $g(z^*)$ coarsens z^*).

Because B_{insuff} is bounded above and B_{closure} below, a positive *separation margin* $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ establishes that the closure’s own error exceeds what input insufficiency can account for, while a non-positive one is a failure to separate and licenses nothing: a reversal would need a *lower* bound on B_{insuff} , which this instrument does not supply.

What population, what is conditioned on, and what sets the resolution. B_{insuff} is well posed only once the population and the conditioning variable are named. The population is the *measurement superpopulation*: a draw is a nominal compound pair *at a temperature*, together with the conformer, tautomer and protonation state actually populated under those conditions, and an independent experimental determination of $\ln \gamma_2^\infty$ for it. The conditioning variable is the closure’s full argument, (z^*, T) where g reads a temperature. Neither the conformational state nor the replicate determination is part of it, so $\text{Var}(m \mid z^*, T)$ is the spread of $\ln \gamma_2^\infty$ across the conformational

ensemble and across replicate determinations at one reference profile and temperature, non-zero for physical reasons unrelated to sample size. Experimental label noise is a *component* of it, not a term beside it: an independent zero-mean error on m enters $\text{Var}(m \mid z^*, T)$ directly, so it is inside every bound we report and never added on. What the design cannot do is *observe* it, neither set holding two rows at a common argument (§S9), so B_{insuff} is reached only from above and B_{closure} only from below. The bounds are conservative in one direction only—were the superpopulation variance negligible the finding would be the stronger $B_{\text{closure}} = \text{MSE}$ —so nothing reported can be an artifact of overestimating B_{insuff} , and no entry is the share of the error the inputs contribute. A low P_{boot} , the cluster-bootstrap frequency that the measured margin stays positive, is likewise loss of resolution and never evidence for the inputs; resolution is governed by $\dim(z^*)/n$.

2.1.2 The decomposition and the grounding gap

The decomposition is the measurement the conclusions rest on: assumption-light, needing neither a representable encoder nor a lossless bottleneck. Alongside it we report a second, purely observable quantity, the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \leq \mathcal{G} \leq B_{\text{closure}}$, so $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem S1, §S2). A1 is empirically doubtful—the encoder is transfer-limited, a linear probe dropping from train $R^2 \approx 0.76$ to test $R^2 \approx 0.45$ —and A2 is unverifiable and physically dubious; when A2 fails, $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency, a well-specified closure with a lossy latent giving $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$ (§S11, and the pKa ortho sweep of §S7). We therefore read $\mathcal{G}$ as corroborating evidence and base the model-dominance conclusion on the decomposition.

The sign rule, and scope. Whether the *trained* physics model outperforms the *trained*

Table 1: The two symbols that carry more than one meaning in this manuscript, and the renamings that separate them.

Symbol	Meaning here
σ	standing alone, the screening-charge-density profile; $\sigma(\cdot)$ with an argument, the generated σ -algebra; σ_H , the Hammett substituent constant of the pKa closure, renamed from σ ; σ_η , experimental label noise
Γ	Γ_S , the COSMO-SAC segment activity coefficient of Eq. (S1); the grounding gap is $\mathcal{G}$, renamed from Γ so that the two do not collide

black box at sample size n follows a sign rule with two halves: a constrained prior helps only where the estimation variance it saves exceeds its asymptotic bias Δ_∞ , and a prior whose asymptotic bias is strictly positive always loses once the budget passes a critical size, the variance it saves vanishing as data accumulate while Δ_∞ does not (Proposition S1, §S2). Each physics-versus-control difference below is a penalty at the budget actually trained, not an asymptotic one; there is no “optimal grounding” theorem, and the decomposition does not improve out-of-distribution accuracy. Everything here is measured at infinite dilution on two sets that differ in regime (§3.2), and transfer of either to the finite-composition regime in which the paradox is observed is a conjecture (§3.6). We read these results against the model-discrepancy, sloppiness, identifiability, misspecification, underspecification and null-space-prior literatures (§S12). Spanning two chemical domains, this paper would otherwise use two letters for more than one quantity each; Table 1 gives the renamings that separate them, and every other symbol is defined where it is used.

2.2 Data, splits, and labels

Solubility measurements come from BigSolDB 2.0:²¹ 127,930 rows over 15,665 solutes and 221 solvents between 243 and 438 K,

partitioned by Bemis–Murcko scaffold into 111,724 training, 8,103 validation and 8,103 test rows so that every test solute is unseen at training; a melting-point-independent miscibility/structure filter sets row membership, and crystal T_m , ΔH_{fus} , Hansen parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. Crystal supervision is scarce—the test and validation splits contain essentially no measured ΔH_{fus} , the quantitative basis of the identifiability argument (§2.1). The external single-component data used for grounding are an open crystal-property pool ($\sim 15,000$ melting points), the VT-2005 σ -profile database, and ThermoML infinite-dilution activity coefficients.²² Two hazards the pipeline handles explicitly—melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions—are recorded in §S9.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule whose scaffold appears in validation or test and aborts on a missing split file — but that is a property of the builder, not a certified property of the runs reported here. One σ -stream build on disk was generated against the wrong split files, 91 of its 1425 rows being held-out solutes by canonical SMILES, and per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm. §S9 gives the file-order evidence, the three facts that bound what a leak could do, and the one molecule it touches in the $n=44$ analysis of §3.3. We therefore do not assert the scaffold guarantee for these runs.

2.3 Model, closure, and training

A *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head; two interchangeable encoder alternatives do not outperform it. A fusion head predicts T_m and ΔH_{fus} , giving the ideal term $\Phi(T)$ of Eq. (1) with the constant- ΔC_p correction omitted (audited in

§S5). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls: the fitted NRTL model and its symmetric one-parameter collapse, in which the identifiability analysis is cleanest (§2.1). The third, parameter-free, is the closure under study—a σ -profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$ with an optional Staverman–Guggenheim combinatorial term, the full/residual-only distinction on which Table 2 and §3.2 turn. That closure over a network-*predicted* σ -profile, and the intervention that swaps it for the external reference, are the object of study (§3.1).

The saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$ as a damped fixed point in x_2 (§S1); a bounded, gated “adaptive physics correction” perturbs the physical parameters rather than the output, so the physical decomposition survives it. External single-component data enter through *grounding streams*—the training-time sense of grounding, a separate operation from the evaluation-time substitution below. Each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor’s mask active), interleaved into training under the guard above; the streams ground the crystal factor from the melting-point pool and the activity factor from the σ -profile database, identifying from external data a factor that solubility alone leaves non-identifiable (§2.1) and that a black-box predictor cannot use. Training runs a three-stage curriculum under a weighted loss: a warm-up on the auxiliary streams alone, then joint training against solubility, then a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of the warm-up, so the σ stream enters the warm-up only (§S3).

The in-house closures, and where their validation stops. The dispersion-off closures in the fidelity measurement (§3.2.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ -averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer—the *fair*

2002 arm wherever the kernels are compared, because it reads all three profile channels—reproduces the NIST reference COSMO-SAC-2002²³ to RMSE 0.0035 $\ln \gamma$ on identical VT-2005 profiles, and our COSMO-SAC-2010/dsp layer,^{24,25} built on the *typed* latent of three donor/acceptor-resolved σ -profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off*. The dispersion term is *not* validated to that standard, which is why every dispersion-off contrast is computed on our layers while the *2010, dispersion on* runs are quoted from cCOSMO. Layer definitions, validation figures and the full scope of the dispersion caveat are in §S1.

2.4 Evaluation protocol, and the DirectGNN control

The *oracle substitution* replaces the predicted σ -profile $\hat{z}$ with the reference VT-2005 profile z^* before the closure, matched by canonical SMILES. It is applied three ways to rule out implementation artifacts: at evaluation only; injected at training time so the model co-adapts; and as a channel-swap control, injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so that only the activity branch refits against it. The last two are at seed 42 only. An analogous override substitutes reference crystal targets. The activity target for the decomposition (§3.2) is the infinite-dilution $\ln \gamma_2^\infty = \lim_{x_2 \rightarrow 0} \ln \gamma_2$, on two sets that differ in their reference database and their regime (§3.2.1–§3.2.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver. The two arms therefore share their representation but not their inputs or their tuning: the control receives an explicit temperature encoding the physics arm does not, seeing T instead through $\Phi(T)$ and the closure, and the optimisation settings were tuned per arm and differ. The TGNNsolv–DirectGNN gap is accordingly a comparison of two separately tuned pipelines that differ in the ther-

modynamic bottleneck, not an isolation of the bottleneck, and every physics-penalty comparison in §3 carries that caveat.

The controlled comparisons of §3.1 are run at three seeds and reported as $\text{mean} \pm \text{sd}$ on a cross-arm row set; 5608 is that lock, the test rows supervised and finite in every arm, so all arms score identical rows. Single-seed, and labelled as such where each is reported, are the training-time and channel-swap oracle controls, the output-residual arms (§3.4), the data-efficiency curve (§§6.5) and the loss-simplification ablation (§S1). The two audit runs behind §3.2 are named there with the script and the artifact file of each; not every artifact those scripts read is small enough to version, and the larger ones are deposited separately (§3.6).

2.5 Estimating the split: one cell, one convention, one rule

Four analyst choices set the law-of-total-variance bound—the bin count, where the equal-count bin edges fall, the within-bin variance convention, and the combinatorial convention—and all four are fixed here, before any margin is reported.

The estimator cell. Eight equal-count bins of $g(z^*)$, the *unbiased* within-bin variance, and the row unit, applied identically to every set and every stratum whatever its size. Enumerating every equal-count contiguous partition, the bin-edge choice is immaterial on the broad IDAC set and material on the VT-2005-matched set, whose margin changes sign inside the envelope; we accordingly read that set’s +0.05 as consistent with zero and carry the claim on the broad set in stratified form. The binning rule and its envelope are in §S3.

The convention rule. The margin depends on the combinatorial convention, so one rule governs both sets: headline the *deployed* residual-only convention, pairing MSE and the bound computed under that same convention. Headlining the deployed convention on the smaller set and the full convention on the broad one would be the favourable choice on each set taken separately; one rule applied to both

costs the broad set’s margin about half its value. The Jensen constant-offset bound is convention-specific—it separates under the full combinatorial convention and inverts under the deployed residual-only default—so the separation reported in §3.2 rests on the model-free law-of-total-variance (LOTV) bound, which fits nothing and so lets no held-out row borrow strength from a near-twin. B_{insuff} is one population quantity while the two conventions coarsen two different statistics of it, so both bounds are valid on it, and so is the smaller. The Jensen numbers are in §S3 and §S2.

The stratification rule. The broad set is stratifiable where the VT-2005-matched set’s 60 pairs are not, on two axes that are pure functions of molecular structure: an ordered nine-class cascade of substructure tests over the solvent, hydrogen-bonding role first and dominant functional group second, and the solute’s role as water or as an organic. The classifier reads a SMILES string and nothing else—never m , g , the squared error, the source or the bootstrap—so no cell can have been chosen on its outcome, which is a property of the code rather than an assurance. A stratum is *boundable* at $n \geq 40$, where eight bins hold five rows; the smaller ones are reported at that same cell rather than at a rule of their own, with the resample-adaptive rule $\max(3, \lfloor n/10 \rfloor)$ carried beside them in the deposited table, nothing being dropped for being small.

The admissibility rule. A stratum is *admissible* only if it is (a) boundable at that fixed cell and (b) robust to leave-one-source-out—its margin keeps its sign when each source publication contributing to it is deleted in turn—in all four unit $\times$ convention cells, since the map reports all four. That last clause is load-bearing and is never dropped: without it the set taken whole passes too. A stratum drawing on one publication fails (b) by non-testability: there is no deletion to perform, so the sign was never put at risk. A deletion leaving fewer than 16 rows leaves the fixed cell undefined and the sign unverifiable, which fails (b) as well. Everything else is *reported in the map and not stated*, with its n , its sources, its margin, its interval and the reason. The rule governs the cells that favour us

and those that do not alike, and it tests whether a number is stable and not what it means: an admissible stratum whose margin is not positive, and an admissible stratum that is another one restated, both establish nothing.

3 Results and discussion

Cluster-bootstrap frequencies P are quoted as bands—rounded to one decimal, with > 0.95 and < 0.2 closing the ends—because with 17 solvent and 41 solute clusters on the VT-2005-matched set they do not carry two-decimal resolution (§S3); the two-decimal values are in the deposited artifacts. Table 2 collects the load-bearing claims with the set each was measured on, its size, and the value with its uncertainty and scope.

3.1 The two senses of grounding

The word “grounding” names two opposite operations on the same VT-2005 database. Supervising the σ head against the database *during training* improves solubility MAE from 2.043 ± 0.040 (ungrounded) to 1.846 ± 0.053 (grounded), a gain of 0.20 over three seeds (Table S2). Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive R^2 (Fig. 2): the model predicts solubility better *from its own learned σ -profile* than when *given the external reference one* (MAE 1.85 vs 2.25, in $\ln x_2$ units). Of the figure’s eight arms two are non-COSMO controls, DirectGNN and a fitted NRTL closure; five COSMO-SAC arms differ only in how the σ head is trained and in what it is fed, and the sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. σ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, *ref. σ (train)* the co-adaptation control injecting it during training instead; per-arm settings are in §S3. The physics arms trail DirectGNN, tuned separately from them, so that gap is not attributable to the bottleneck (§2.4).

The headline $+0.41$ is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. The control that removes that component injects the reference profile *at training*, so the model co-adapts; at seed 42 it costs $+0.18$ ($1.803 \rightarrow 1.981$) where the evaluation-only substitution costs $+0.43$ at that same seed ($1.803 \rightarrow 2.232$). One seed does not establish it, but the direction points to part of the headline gap going with co-adaptation; the co-adapted arm is by design the confound-free one. A channel-swap control costs $+0.27$, to 2.08. Both are worse than the learned- σ arm, which rules out an implementation artifact, and both are single-seed, so they fix a sign and not an effect size, as does the 0.59 share they imply. Only the three-seed arms can be set against one another, and there the substitution costs about twice what supervision gains ($+0.41$ against -0.20); the co-adapted arm’s $+0.18$ cannot enter that comparison. The between-seed arithmetic that excludes it, and the row set the three oracle controls are computed on, are in §S3.

Which of the two causes the substitution fails from is not decided here. Drug-like solutes are essentially absent from the VT-2005 set, so the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers only ≈ 7 solutes—too few either way, and enlarging that set would mean computing new reference profiles rather than looking them up. The causal claim is therefore carried on the activity axis, where the reference input is clean and matched on both channels: the decomposition (§3.2) and the latent’s departure from its reference ($n=44$, §3.3). There the fixed reference-input closure sits well above the bound on what its inputs can fail to resolve, under-predicting $\ln \gamma_2^\infty$ with a systematic bias (Fig. S3).

3.2 Closure against inputs on the activity axis

The externally-referenced decomposition of §2.1.1 separates a fixed closure’s error into a part in the map and a part in its inputs. Its

Table 2: One row per load-bearing claim: what is claimed, the data set and its size, and the value measured with its uncertainty and its scope. Notes *a–k* qualify a row or a column.

Claim	Set (<i>n</i>) ^a	Value, with its uncertainty and scope ^b
<i>The two senses of grounding</i> (§3.1)		
Reference profiles hurt solubility at evaluation (MAE)	test split (5608), 3 seeds	1.85 → 2.25; the per-seed values do not overlap
...and still hurt when injected at training	cross-arm rows ^c , seed 42	1.80 → 1.98 (own baseline 1.80 → 2.23); one seed, so a sign and not an effect size
...so part of that gap is test-time distribution shift	cross-arm rows ^c , seed 42	share 0.59 on one seed: a description of that seed, not a magnitude
Training-time σ supervision helps (MAE)	cross-arm rows ^c , 3 seeds	2.04 → 1.85; the per-seed values do not overlap
<i>Closure against inputs on the activity axis</i> (§3.2)		
$B_{\text{closure}} > B_{\text{insuff}}$ on glycol-ether solvents ^{d,i,k}	broad IDAC, 182, 3 sources	+2.04 [+1.27, +2.83] to +2.94, $P > 0.95$ in all four cells; worst deletion +1.60
... on every other stratum of the map	broad IDAC, 59 strata	Tables 3–4 and §S3.1, each with its margin, interval and the test it fails; none is stated as a finding ^k
... for either set as a whole	broad IDAC (477); VT-2005-matched (60)	+0.51 (full +0.95); −0.38 compounding two cuts (+0.01 full); +0.18 less one publication; +0.05 on the smaller set: an average weighted by composition ^j
$B_{\text{insuff}}^{\text{up}}$ near stratum-independent, MSE not	broad IDAC, 3 boundable classes; 6 with an estimate	×2.0 against ×25; ×6.8 against ×51; $p=0.02$ against 2000 null partitions of 9 classes
$B_{\text{closure}} > B_{\text{insuff}}$, 2010 on its own typed latent ^j	broad IDAC (477); VT-2005-matched (60)	+1.29, $P > 0.95/\approx 0.9/\approx 0.9^e$; −0.77 at $P \approx 0.3$ on the smaller set, where the ordering is lost: loss of resolution, not a reversal
<i>Whether a better closure moves the ceiling</i> (§3.2.3)		
The 2002 → 2010 step lowers the error	broad IDAC (477)	−0.99 at $P \approx 0.2^f$: no improvement, and not a measured equality either
...but does on the VT-2005 set’s chemistry ^g	57 shared pairs, UD	1.78 → 0.77, $P > 0.95$; those 57 pairs are also broad-set rows, so not an independent sample
A fitted kernel residual transfers across solvents	VT-2005-matched (60), grouped folds	−5 ± 16%: the interval spans zero, and excludes a cross-solvent reduction above about 30%
<i>Mechanism</i> (§3.3)		
The latent departs from its reference along a shared direction	matched molecules (44), 3 seeds	51 ± 2%; the magnitude is not stable across analysis configurations
...and that departure’s top-two mode share exceeds a phase-randomised null	one grounded checkpoint, same 44	0.40 against 0.22 ± 0.01 ^h ; one checkpoint, and a wrong-molecule null is not cleared
...and that direction transfers to molecules outside the fit	matched molecules (44), 3 seeds	4.1 ± 0.5 × a zero-drift null, with no constant-offset control on these seeds; the ± spans seeds at one partition
That departure cancels closure error	not measured	the one direct check is under-powered and runs against it
<i>Accuracy</i> (§3.6)		
Physics arm vs. DirectGNN accuracy (ΔMAE)	test split (5608)	+0.14 between separately tuned pipelines, so not attributable to the thermodynamic bottleneck

^a Broad IDAC, the 477 IDAC ∩ UD measurements over 185 distinct pairs at all temperatures; VT-2005-matched, the $n=60$ pairs at 298 K carrying a VT-2005 profile; UD, the University of Delaware σ -profile database. Resolution $\dim(z^*, T)/n$ per set: §3.2.1–§3.2.2.

^b Margin is $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$, positive when B_{closure} exceeds B_{insuff} ; P is a two-way (solute × solvent) cluster-bootstrap frequency, reported as a band, by the rounding rule at the head of this section. ^c The cross-arm row set common to every arm of Table S2.

^d LOTV rows use eight equal-count bins, the unbiased within-bin variance, and the *deployed residual-only* convention on both sets and both kernels (§2.5). ^e Bootstrap band over pair / two-way / source clusters. ^f Deployed convention, dispersion off; under the full convention −0.07, $P \approx 0.6$, and +0.06, $P \approx 0.6$ with dispersion on (Table S6). ^g At fixed profile database and temperature, full convention. ^h ± is the null’s spread; the fine-tuning drift has no matched null of its own. ⁱ One estimator cell for every stratum, under a rule reading molecular structure only (§2.5). B_{insuff} is bounded above and B_{closure} below, so a positive margin establishes that the closure’s own error exceeds what input insufficiency can account for; a non-positive one is failure to separate and licenses nothing, a reversal needing a lower bound on B_{insuff} this instrument cannot give. ^j An aggregate margin is a composition-weighted average, and the deletions and cuts recorded here are failures of the headline cell rather than robustness passes (§3.2.1). These rows record what the aggregate was and how it fails; the rows above replace it. ^k Admissible (§2.5, applied alike to every stratum): boundable at the fixed cell and keeping the margin’s sign when each contributing publication is deleted, in all four unit × convention cells. Of fifty-nine strata, fifteen are boundable and six admissible, of which one row set is also positive and not contained in another—the glycol ethers, the one row set stated as a finding. Every boundable stratum is in Tables 3–4 and the rest in §S3.1, each stating what it shows or the test it fails.

The grounding paradox: reference σ -profiles give the worst arm

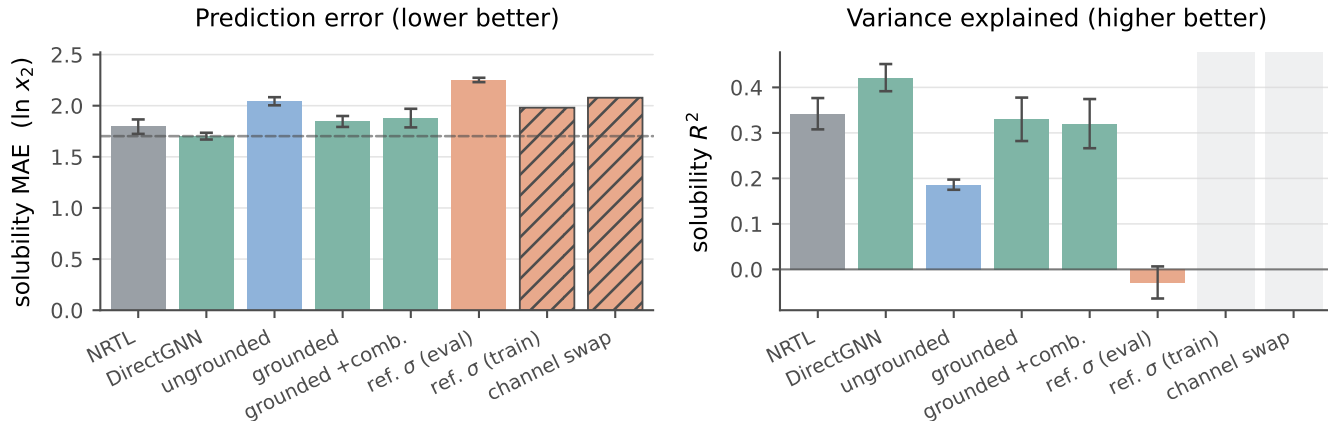


Figure 2: Controlled comparison on the scaffold split ($n=5608$). Solid bars: three seeds, mean $\pm$ a population standard deviation (ddof= 0). The two hatched bars are the co-adaptation controls, seed 42 only, so they carry no error bar and no R^2 , and keep an empty slot in the right panel. Dashed in the left panel only, the DirectGNN reference.

output is a map over chemical strata rather than one verdict for a set, for the reason §3.2.1 establishes. It runs on two data sets that must not be conflated—the *broad IDAC set* ($n=477$, UD profiles, all temperatures) and the *VT-2005-matched set* ($n=60$ at 298 K, where the paradox’s own database supplies both profiles)—and then asks whether a better closure moves the ceiling.

3.2.1 The broad IDAC set ($n=477$)

The closure reads a temperature as well as a profile pair and is evaluated at each row’s own temperature, so the conditioning variable is its full argument (z^*, T)—the concatenated UD profile pair with T , 103 dimensions against $n=477$ (dim / $n \approx 0.22$)—as §2.1.1 requires, and the target m is the experimental $\ln \gamma_2^\infty$, so floor and closure error are measured on one data set and one profile database. Its chemistry is small: no solute exceeds 12 heavy atoms (median 7), against a median of 14 and a maximum of 122 for the drug-like solids of this paper’s own solubility corpus, and only 0.46% of corpus rows carry a solute from this set. *Broad* therefore names the reach of the γ^∞ corpus the kernel is calibrated and benchmarked on and not coverage of the chemistry a solubility model is de-

ployed against: it covers the solvent side of deployment and not the solute side. It is matched to UD by InChIKey with a first-block fallback, a looser rule than the VT-2005 set’s; a wrong z^* inflates B_{closure} , so the looser rule runs against the ordering we claim, and the aggregate margin is unchanged on the 465 exactly matched rows (§S9).

Under the convention rule of §2.5 the fair 2002 arm scores $\text{MSE} = 1.902$ against $\text{Var}(m) = 4.85$, the LOTV bound gives $B_{\text{insuff}} \lesssim 0.70$ and hence $B_{\text{closure}} \gtrsim 1.21$: a separation margin of +0.51 for the set taken whole, +0.95 under the full convention and +0.27 collapsed to one row per distinct pair, positive in every cell of the bin-count $\times$ variance $\times$ convention grid, with two of the three bootstrap clusterings extending just below zero. Both sweeps, and the tighter-of-the-two-conventions reading we do not headline, are in §S3.

That aggregate is not the instrument’s output. Two operations reported separately—the pair unit, and dropping the two chemistries the error concentrates in, the glycol-ether solvents and the rows carrying water as the solute—compound badly. The chemistry cut alone takes +0.51 to -0.12 on the 258 rows that remain, the pair unit alone gives +0.27, and the two together give -0.38 on 136 pairs; under

the full convention the same pair of cuts takes +0.95 to +0.28 to +0.01. Nor is the aggregate carried by the set: deleting one source publication at a time, 10.1021/acs.jced.7b00114—37 rows, six molecule pairs, 32% of the set’s squared error—takes the margin to +0.18, while each of the other fourteen deletions leaves it between +0.42 and +0.87 (under the full convention that curve is flat, [+0.82, +1.07] against +0.95). That single deletion loses two thirds of the margin and its resolution without reversing its sign; the sign reverses only under the compounded cut. Both are failures of the headline cell rather than robustness passes, and the aggregate margin is an average weighted by one set’s chemical composition. Two further caveats qualify the corroborating estimators and not the bound. Under random folds they leak, and under folds blocked on molecules they are uninformative on the VT-2005-matched set and split on the broad set, so on neither set do they support the separation; and the binning-free pair bound, tighter though it is, is not promoted, because its multi-row pairs each come from a *single* publication and it therefore samples no between-laboratory component. Read together, the two say the binning bound is slack rather than that B_{insuff} is negligible. What is reported rests on the LOTV bound, which fits no model and so has no fold design to leak through (§S3, Table S7).

The map, and how much of it survives the rule. The map runs over 59 strata of the broad IDAC set—477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications—on the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole. Tables 3 and 4 are the fifteen the fixed estimator can bound, the only ones whose margins the rule can act on; the other 44, including the 24 that carry no estimate at all, are in §S3.1 with the test each fails, drawn there (Fig. S2) and with their error, bound and error share (Tables S8–S12). Six of the fifteen are *admissible*—boundable and sign-stable under deletion of each contributing publication, in all four unit $\times$ convention cells. Those six are three distinct row sets, of which one is both positive and not contained in another: the glycol-ether sol-

vents. **That row set is the one established finding:** on 182 rows over 43 pairs from three publications carrying 45% of the set’s squared error, the closure’s own error exceeds what input insufficiency can account for, by +2.04 with a 90% interval of [+1.27, +2.83] at the deployed convention—+2.94 at the pair unit under the full one—positive in all four cells and under every single-publication deletion, the worst leaving +1.60 when 10.1016/j.jct.2016.10.013, which carries 53% of that stratum’s squared error, is removed. It is not an artifact of one solute family: the 74 rows left when its alkane solutes are dropped keep the sign at +1.40 deployed and +1.69 full, under all three deletions.

Three of the six are that same row set at three granularities. The remaining three are two row sets, each carrying its reason in Tables 3–4. The alkane solutes (129 rows, 45 pairs, +1.38, [+0.79, +2.33]) are 84% the glycol-ether rows and return −0.22 on the 21 outside them—itsself a sub-threshold number, so the demotion rests on the overlap arithmetic and not on it. The acceptor-only aprotics (57 rows, −0.19, negative in all four cells, $P < 0.2$ at both units under the deployed convention and $P \approx 0.3$ under the full one, at an MSE of 0.089 that is at the scale of the measurements) occupy two cells, pass the rule and license nothing, the instrument being one-sided; their deletion test is additionally *anti-conservative*, two of the four deletions leaving $n < 40$, where the Bessel within-bin variance inflates—0.140 to 0.258 on the $n=38$ remainder—and drives a negative margin further negative. That is the whole of the uniqueness claimed for the glycol ethers: the only row set admissible, positive and not contained in another—not the only admissible stratum and not the only one sign-stable under every deletion.

Every other stratum is reported and not established, each with the test it fails. Water carried as the dilute solute (37 rows, +6.33) and the halogenated solvents (30 rows, +3.39) are too small to bound, one publication supplies both and its deletion cannot be run, so those two large margins were never put at risk; the alkane solutes and the acceptor-only aprotics pass the rule and establish nothing, for the

Table 3: **The map, solvent axis: every solvent stratum the estimator can bound.** One cell applied identically to all of them—eight equal-count bins of $g(z^*)$, unbiased within-bin variance, row unit, deployed residual-only convention; *full* is the same cell under the full combinatorial convention. With Table 4 these are the 15 of the map’s 59 strata with $n \geq 40$, where eight bins hold five rows each; the 44 smaller ones, 15 of them with no estimate at any binning, are in §S3.1 with the test each fails and with every stratum’s error, bound and error share. Nothing is dropped from either place, and the stratification reads molecular structure alone, so no cell was selected on its outcome.

Stratum	n	pairs	sources ^a	margin	90% margin ^b	full	cells ^c	what it shows, or the test it fails ^d
<i>Solvent class</i> glycol ether	182	43	3 (P8, 0.53)	+2.04	[+1.27, +2.83]	+2.23	4	the closure’s own error exceeds the bound on its inputs , and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 $\rightarrow$ −0.075; P14 takes +0.046 $\rightarrow$ −0.194
aprotic acceptor	57	57	4 (P7, 0.32)	−0.19	[−0.44, +0.12]	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
<i>Solvent class, coarse</i> H-bond donor and acceptor (O)	328	75	8 (P8, 0.40)	+0.53	[−0.21, +2.18]	+1.34	3	the margin changes sign when one publication is deleted: P15 takes +0.527 $\rightarrow$ −0.417
H-bond acceptor only	70	59	5 (P13, 0.96)	+1.43	[−0.28, +6.19]	+0.77	0	the margin changes sign when one publication is deleted: P13 takes +1.434 $\rightarrow$ −0.192
<i>Solvent family, fine</i> glycol ether	182	43	3 (P8, 0.53)	+2.04	[+1.27, +2.87]	+2.23	4	the closure’s own error exceeds the bound on its inputs , and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	[−0.16, +0.40]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 $\rightarrow$ −0.075; P14 takes +0.046 $\rightarrow$ −0.194

^a Source publications, then the largest and its share of that stratum’s squared error. The keys, used in both documents:

P1 10.1016/j.fluid.2013.06.028, P2 10.1016/j.fluid.2015.01.010, P3 10.1016/j.fluid.2015.11.037, P4 10.1016/j.fluid.2016.08.030, P5 10.1016/j.fluid.2016.10.009, P6 10.1016/j.jct.2013.05.006, P7 10.1016/j.jct.2015.12.034, P8 10.1016/j.jct.2016.10.013, P9 10.1016/j.jct.2018.05.003, P10 10.1016/j.jct.2019.05.011, P11 10.1016/j.tca.2012.02.005, P12 10.1016/j.tca.2015.11.010, P13 10.1021/acs.jced.7b00114, P14 10.1021/acs.jced.9b00726, P15 10.1021/je020196d. ^b Two-way solute $\times$ solvent cluster bootstrap, 3000 draws, redrawn at each granularity, so one row set’s

margin repeats while its endpoints differ in the last digits. ^c In how many of the four unit $\times$ convention cells the stratum is boundable and keeps its margin’s sign under deletion of each contributing publication; the rule (§2.5) requires all four. ^d The condition the row is in, not a label for it. A margin is one-sided: positive establishes $B_{\text{closure}} > B_{\text{insuff}}$ there, non-positive is failure to separate and licenses nothing (Table 2, note *i*). Rows at different granularities, and on the two axes, can be the same measurements, so each row that keeps its sign carries its overlap and no row set is counted twice.

Table 4: **The map, solute axis, the solvent $\times$ solute cross-tab, and the set as a whole**—the second half of Table 3, at the same estimator cell and under the same rule, whose notes apply here unchanged. The whole-set row is the aggregate §3.2.1 declines to state, printed with the rest.

Stratum	<i>n</i>	pairs	sources ^a	margin	90% margin ^b	full	cells ^c	what it shows, or the test it fails ^d
<i>Solute role</i> organic solute	440	179	14 (P8, 0.35)	+0.18	[−0.23, +1.36]	+0.82	2	the margin changes sign when one publication is deleted: P15 takes +0.184 $\rightarrow$ −0.094
<i>Solute family</i> alkane	129	45	7 (P15, 0.53)	+1.38	[+0.79, +2.33]	+1.72	4	the sign holds under every deletion, but 84% of these rows are the glycol ethers’ and the 21 outside give −0.22: these are those rows again; 10% shared with the aprotic acceptors
aromatic	71	32	7 (P15, 0.53)	−0.19	[−1.01, +0.75]	−0.07	0	the margin changes sign when one publication is deleted: P6 takes −0.188 $\rightarrow$ +0.081
hydrocarbon	63	11	3 (P14, 1.00)	−0.53	[−0.85, +0.64]	−0.33	0	too few rows remain after deleting P14 (1 of 3 deletions), so the sign is untested there
<i>Solvent class $\times$ solute role</i> glycol ether organic solute	182	43	3 (P8, 0.53)	+2.04	[+1.23, +2.90]	+2.23	4	the closure’s own error exceeds the bound on its inputs , and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water organic solute	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 $\rightarrow$ −0.075; P14 takes +0.046 $\rightarrow$ −0.194
aprotic acceptor organic solute	57	57	4 (P7, 0.32)	−0.19	[−0.40, +0.12]	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
<i>The set as a whole</i> whole set	477	185	15 (P13, 0.32)	+0.51	[−0.09, +2.16]	+0.95	3	passes here and fails at the pair unit (deployed convention), where deleting P13 takes +0.270 $\rightarrow$ −0.119

reasons above; the coarse acceptor-only class is carried by one publication’s aryl-ether rows and turns negative without them; water as a solvent, the coarse hydroxylic class and the organic-solute subset each lose their sign under a deletion, which is why the finding belongs to the glycol-ether sub-class and not to the hydroxylic solvents as a class; the mono-alcohols, the N–H protics and the ester and azine sub-families, which an acceptor-only reading would lean on, are unboundable with a deletion that cannot be run. None may be used to explain a reversal. The whole-set row passes at the headline cell and fails once the pair unit is required, so the rule does not by itself dispose of the aggregate there; the composition argument above does.

Why the output is a map. $B_{\text{insuff}}^{\text{up}}$ is nearly stratum-independent and MSE is not. Across the three classes the design can bound—glycol ether, water and the acceptor-only aprotics, bounds 0.108, 0.215 and 0.140—the bound spans a factor of 2.0 against errors (2.252,

0.476, 0.089) spanning a factor of 25; adding the three that carry an estimate without being boundable—mono-alcohol 0.733, halogenated 0.592, N–H protic 0.281—gives 6.8 against 51, that 6.8 being set by the mono-alcohols’ 0.733, which exceeds that stratum’s own $\text{Var}(m) = 0.604$ and is flagged in the artifact as an invalid bound on a stratum the boundability criterion has already excluded. The two orderings are unrelated—the glycol ethers carry the lowest bound of any class and the second-highest error—and against 2000 random partitions into blocks of the observed class sizes, at the same cell, the chemical partition’s spread in MSE exceeds every draw while its spread in the bound is unremarkable ($p = 0.26$); their ratio gives $p = 0.02$. That is the estimator showing through: every stratum is summarised by eight quantile bins of one scalar, so the bound measures the residual spread of m inside such a bin, a property of the binning rather than of the chemistry, while nothing constrains the error to follow it. An aggregate margin is therefore

a composition-weighted average of a term free to move 25-fold offset against one that barely moves, and it will report a binding closure for any set whose composition happens to include a chemistry the closure fails on—a statement about the aggregate and not about the bound, which stays valid in each stratum where it is computed.

3.2.2 The VT-2005-matched set ($n=60$)

This set carries the paper’s other reference-matched analyses, and it is far weaker. All 60 rows sit at 298 K, so z^* is the profile pair alone and $\dim(z^*)/n \approx 102/60$ leaves the split behind one-sided bounds; it is strongly unbalanced, methanol occupying 25 of the 60 rows and the two worst-fit pairs carrying 34% of the squared error; and its 17 solvents fall in six of the nine classes, none reaching the $n=40$ a bound needs, so it supports no stratified bound and no stratum of it is stated as a finding. Its aggregate margin is +0.05, and it does not stand: deleting the two leverage pairs takes it to -0.002 , dropping methanol loses it, blocked out-of-fold estimators do not support it, and 11 of the 60 single-pair deletions take it non-positive, where on the broad set no single-pair deletion moves the sign at all. Zero-mean label noise can understate a separation but not create it (§2.1.1); what a margin does bound is a label systematic tracking z^* , implausible on the maximum-likelihood binning and *not excluded* on the Bessel-corrected one. Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are not independent samples—this is the sub-design VT-2005, and therefore the paradox’s own oracle, is defined on, and the agreement between the two sets’ error orderings is a consistency check rather than corroboration. The composition audit, the stratified error ordering, the four fold designs, the cell grids, the cluster bootstraps, the label-noise arithmetic and the convention, leakage and γ -stratification stress-tests are in §S3 and §S3.1.

3.2.3 Whether a better closure moves the ceiling

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 $\rightarrow$ 2010 step delivers one on the VT-2005-matched set ($n=60$) and not on the broad IDAC set ($n=477$), and a block split identifies the factor that carries that reversal.

The two kernels have no *prescribed* common input, σ -profile averaging being part of each model’s definition, so evaluating each on its native profile is the specified use. An input-fixed *control*—profile database, cavity area, volume and averaging held fixed, kernel alone varied—gives $2.047 \rightarrow 0.765$ on the smaller set ($P > 0.95$), under three limits: it cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction; its step is larger than the native-input one, so it is favourable to the conclusion it supports; and its re-averaging term is unresolved. That arithmetic, the NIST implementation detail fixing the *fair 2002* arm at 1.757, and the three arms—*fair 2002*, *2010*, *dispersion off, 2010*, *dispersion on*—on both sets under both conventions are in §S3 and Table S6.

The step more than halves the smaller set’s error under the full convention ($P > 0.95$), the convention under which the two kernels’ published error figures are reported, and cuts it 42% under the deployed one. On the broad IDAC set it is not an improvement under either rule, and the 3% the dispersion term recovers there is a null whose interval is an order of magnitude wider than the effect, not a measured equality. That arm’s run on the smaller set is on $n=57$, three halogenated pairs having no tabulated dispersion value—a coverage limit of the model offered as the lever; no claim rests on the 0.765-versus-0.785 ordering, which §S3 shows is not an artifact of the differing n .

The reversal is the *chemistry* and neither the profile database nor the temperature range: with UD profiles, one extraction and one convention throughout, the *2010, dispersion off* arm wins on the 57 pairs shared with the smaller set, does not win on the complemen-

tary 420, and still does not win on the 80 complementary rows that are *also* at 298 K—that last block at $P \approx 0.3$ and no interval, the three blocks not having been tested against one another (Table S6). The VT-2005-matched set is a low- γ , hydrogen-bond-dominated sub-design, the regime the donor/acceptor-typed 2010 kernel was built for. The map does not corroborate the reversal from the other side and we do not ask it to: the typed kernel is not an improvement on the broad set, of whose squared error the glycol ethers carry 45%, and the classes that would explain a reversal—the halogenated solvents, the aryl ethers and water as the solute, largely one row set, all 13 aryl-ether rows and 18 of the 30 halogenated being water-asolute rows from one publication—are the cells the rule excludes. The block split is therefore the whole of the evidence for the reversal being chemical.

The floor checks are same-database, same-latent *and* same-convention. The *UD-native* check on the smaller set, the one commensurable with the broad set, is at zero under the deployed rule and +0.69 under the full convention. Under the 2010 kernel’s own typed latent, 306-dimensional and so several times worse resolved, the ordering is *lost* on the smaller set (−0.77) and holds on the broad set (+1.29)—both aggregates, and read as such. That is loss of resolution, not a demonstrated flip to the inputs, which would need a *lower* bound on B_{insuff} (§2.1.1); the arithmetic under both conventions and all three clusterings is in §S3.

We know of no large-scale infinite-dilution benchmark of the 2002 $\rightarrow$ 2010 step, and our search was not exhaustive, so this is an absence of evidence rather than evidence of absence; the broad-set null lies within the range of magnitudes the comprehensive assessment²⁶ reports. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set’s chemistry**, and that null is additionally a small-molecule and glycol-heavy one (composition audit, §S9), so on drug-like solids the lever is untested rather than absent.

3.2.4 Correctability, and where the closure fails chemically

Where B_{closure} dominates—on this set, the one row set the rule establishes—the ceiling is in the map; that does not say the map is *fixably* wrong. A small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel, fit on the reference VT-2005 profiles, removes $46 \pm 5\%$ of the reference-input error under random folds but $-5 \pm 16\%$ once folds are grouped so that no solvent is shared. Only the grouped design is free of leakage, the closure error being predominantly a between-solvent systematic. Stated as a bound rather than a null, that design excludes a cross-solvent reduction above about 30%. B_{closure} is therefore a defect a handful of kernel parameters can absorb *within* a solvent and that we have not shown to be correctable across solvents—weaker than a general repair.

Where it fails is chemically placed, and placed differently on the two sets. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents and the closure is near-exact on acceptor-only ones, putting that set’s B_{closure} in COSMO-SAC’s documented single-parameter H-bond kernel—the specific defect the donor/acceptor-typed 2010 closure was built to fix. On the broad set that reading does not reproduce, and neither does its replacement, an absent dispersion term: which of the two the broad set points at is decided in either direction by the same publication the map already declines to state, so we state neither, and the H-bond reading remains the VT-2005-matched set’s, neither corroborated nor replaced by the broad set. That arithmetic is in §S3.1. The residual’s fold design, rank ordering and the scope of its standard errors, its failure to gain over the free 2002 $\rightarrow$ 2010 step above, the NIST reproduction that rules out an implementation artifact, and the Staverman–Guggenheim counter-test in which the physically *more* complete combinatorial term *increases* error on size-asymmetric pairs are in §S3.

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualifi-

cation of the result: it is what a practitioner adopts by default, a closure’s fidelity is *not knowable in advance* on a new system, and the field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. What follows when it is not is that the learned latent silently departs from the quantity it is named after (§3.3), invisibly without an external oracle. Outside that one row set the map reports cells it cannot resolve, not cells in which the closure is sound.

What B_{closure} is a property of. The external ground truth used to charge B_{closure} against B_{insuff} is quantum chemistry, not experiment: the VT-2005 profiles are computed for a single reference conformer and protonation state (§S9), and the successor Delaware database revises those conformations for about a third of its compounds.²⁶ That gap between z^* and the state actually populated under the measurement does not load on B_{insuff} alone: if the reference is displaced, even a perfectly specified g carries a term in B_{closure} , whose share we do not bound. B_{closure} as measured here is therefore the misspecification of a composite object, the kernel applied to one fixed computed conformer, and not of the kernel at the true conformational ensemble—but the deployed pipeline is that same composite, so this is the quantity a practitioner faces. Why the superpopulation has a non-degenerate B_{insuff} at all, and the conformer term’s own arithmetic, are in §S9.

3.3 The end-to-end latent against its reference

A candidate *mechanism* for the low-fidelity, end-to-end regime is this: a σ -profile head trained *end-to-end* through a fixed, misspecified closure—with no σ supervision in that phase—is under pressure to *cancel* the closure’s error rather than to be physically faithful. The hypothesis has two parts and this section establishes only the first. Its antecedent—that the latent leaves the reference along a direction shared across molecules rather than as per-molecule noise—is measured. Its consequent—that the departure cancels part of B_{closure} —is

not, and the one direct check we can run on retained checkpoints argues against it, on the wrong axis and with little power. The external reference that fixes the ceiling (§3.2) makes the first part testable: we compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile molecule by molecule ($n=44$).

The drift under fine-tuning. The comparison is within a single grounded model, checkpointed before and after solubility fine-tuning (the σ head unfrozen), so architecture and initialisation are held fixed and only the objective differs; it is repeated over *three seeds* and reported mean $\pm$ sd. Under σ supervision the head reproduces the reference profile to within $36 \pm 1\%$, so the intermediate *can* carry the physical shape, though only as an in-sample fit: all 44 probe molecules lie inside the σ -grounding pool (§S5). Solubility fine-tuning then drives the profile $41 \pm 3\%$ off the grounded one, for a total departure from the reference of $51 \pm 2\%$. What this section carries is the sign of the departure and the presence of a shared direction, not the magnitudes: that figure, and the two below it, are not stable across analysis configurations (§S5). The drift is not per-molecule noise, and its structure is not a constant offset: the first two principal components of the *mean-centred* deviation carry $72 \pm 1\%$ of the variance. The quantity read against a null is a different one, the top-two share of the learned-versus-reference deviation, 0.399; its null is *phase-randomised*, matching each deviation’s own smoothness and destroying only the between-molecule alignment at issue, and the observed share clears it (0.399 against 0.221 , $p < 0.001$), at roughly $1.8\times$. The $72 \pm 1\%$ mean-centred share has no matched null of its own, a wrong-molecule null built from real reference profiles does *not* clear (0.772 , $p = 1.00$), and if the case for preferring the phase-randomised null is not accepted the spectrum is uninformative rather than supporting (§S5). To test whether the structure transfers across molecules we fit one distortion operator D on half the 44 molecules and score it on the other half, where it predicts the drift $4.1 \pm 0.5\times$ better than the zero-drift (identity) null out of sample.

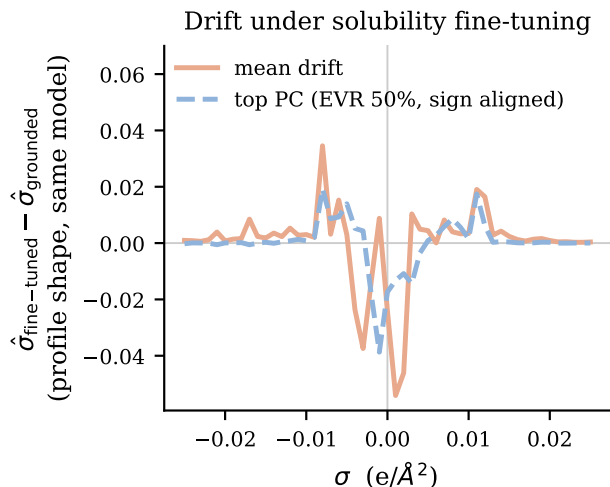


Figure 3: The drift solubility fine-tuning imposes on the learned σ -profile ($n=44$ VT-2005-matched molecules): $\hat{\sigma}$ after fine-tuning minus the *same model’s grounded profile*, not minus the external reference. Shown are the mean drift and its dominant PCA mode, defined only up to sign and drawn in the mean drift’s direction; the legend’s EVR is that mode’s explained variance ratio.

That null is deliberately weak, nothing controls it on these three seeds, whose checkpoints were not retained, and a molecule-independent constant offset reaches $2.1\times$ on the one companion run where that control can be computed, so the ratio stands against the identity null alone; the ± 0.5 spans seeds at one partition, and the held-out half is homologue-saturated. Structure *beyond* an offset therefore rests on the mean-centred spectrum, not on the ratio—and that spectrum does not exclude a shrinkage of every profile toward the corpus-mean shape. A reference ladder calibrating the 51% against uninformative baselines is in §S5; on it the drift consumes essentially all the molecular specificity grounding had provided, as a collapse toward the corpus-mean shape would also look.

Dominant mode, and the direct cancellation test. The dominant drift mode (Fig. 3) transfers area *out of* the nonpolar bins ($\sigma \approx 0$) and *into* the polar wings: the network represents molecules as more polar than they are. Whether that shift *cancels* part

of B_{closure} —the input compensating the map, as the non-identifiability of the split allows (Lemma S1)—is the comparison $\text{MSE}(m, g(\hat{\sigma}))$ against $\text{MSE}(m, g(z^*))$ on the same pairs. It does not support the cancellation reading. Holding the cavity area at its reference value so that only the profile shape varies, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint gives 18.75 against 1.47: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure, not better. Three limits sit on that number. The checkpoint is the uncontrolled one, whose departure (82%) is half again the three-seed $51 \pm 2\%$, so it may not be representative of the three seeds. The axis is wrong: the drift is produced by the finite-composition solubility objective and scored here at infinite dilution. The check also has almost no power in the direction that matters, so it can separate “no cancellation” from “large cancellation” but not from “some cancellation” (§S5). It is the only direct evidence available, and it is evidence against rather than a clean pass or fail. A second, independent check on the closure’s own axis points the same way: a *more* polar profile moves g further from m (§S5). Both sit on that same infinite-dilution axis, so neither refutes cancellation in the regime that generates the drift; they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. What is established is weaker and still consequential: the model’s “physical” intermediate is not the molecule’s charge distribution, consistent with cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification,^{20,27} and that adapting a representation can itself collapse an otherwise physically-faithful latent.²⁸ Its *structure*—low-rank rather than diffuse—excludes a random collapse but not the structured shrinkage above; the diagnostic that would separate them is under-resolved here (§S6.6).

What leaves σ under-determined by activity. The freedom that would make such a cancellation possible is the closure’s geometry: at infinite dilution the activity map con-

centrates on the one or two hydrogen-bonding directions where the closure is least faithful (§S6.4). The end-to-end head can therefore move σ along the remaining, low-sensitivity directions at little cost to the fit—the σ -profile form of Lemma S1. The Fisher measurement behind that statement, and the two respects in which it does not settle the question, are in §S5; nothing here shows recovery of the physical profile to be impossible in principle. The sign of grounding turns on *both* axes, and on which sense of grounding is meant. Freezing the σ head at its supervised warm-up state instead of letting it drift end to end cuts the reference-input penalty by 65% (Δ MAE from +0.40 to +0.14 over $n=5571$ matched test rows). Those are a separate matched pair of runs, one base resumed twice with the freeze off and on, and not the +0.41 headline arms of §3.1 (§S3). A near-physical intermediate makes the pipeline far more robust to substituting the reference σ . Supervision alone does not, however, *flip* the sign here—the penalty stays positive, the closure still being the low-fidelity COSMO-SAC-2002 and the warm-up σ itself only partly physical. The full flip needs supervision *and* fidelity together, which is what TeNNet-SAC⁷ has and this pipeline does not (§1).

3.4 A downstream correction on the output

If the closure binding were merely an inaccurate point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not: an additive bounded residual with a learned confidence gate leaves the gate effectively shut, and forcing it open removes the systematic *bias* (+0.45 $\rightarrow$ +0.06) but recovers little *accuracy* (R^2 0.236 $\rightarrow$ 0.274 on the same rows), as expected if the closure, not the estimate, is the binding constraint. Both arms are seed 42 only, so this fixes a sign and not an effect size, and it is a statement about the one correction tested: an additive bounded residual on the output, gated, on this closure and this corpus. Other downstream corrections, and this one at other seeds, are untested here.

3.5 A second chemical closure: pKa through a fixed Hammett relation

The instrument is not specific to COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. There the fixed closure is a Hammett linear free-energy relationship in a substituent constant, the learned intermediate is that constant predicted from structure, the external reference is the Hansch–Leo table, and the data are the OPERA experimental compilation, split within scaffold $K=20$ times. On the meta- and para-substituted aromatics the relation was built for ($n=158$), substituting the tabulated constant for the learned one *improves* accuracy—MAE 0.48 against 0.315, the reference better in 19 of 20 splits and in three of the four reaction series; on the ortho-substituted and heteroaromatic ones it does not describe ($n=846$), the same substitution *degrades* accuracy—0.83 against 1.62, the reference worse in 20 of 20 splits and in all four series. The sign of grounding therefore reverses inside one closure, in the direction the penalty rule (Proposition S1) gives a priori and on the channel it names: the ortho and steric information the tabulated constant omits, which is input insufficiency, rather than the closure-error cancellation of §3.3. Only $G=4$ Hammett reaction series exist, so the bootstrap resolves nothing finer than series-level sign agreement, read as a sign test on four units ($p=1/16$ on the pole where grounding hurts); the construction, the estimator validation, the measured fidelity sweep and the crossover law that predicts both signs are in §S7.

3.6 Scope of the results

The result is a cautionary measurement about one of the two operations the word “grounding” names. Supervising a learned physical latent on reference values during training helps; *supplying* those values as an input at inference hurts, and it hurts because the fixed map that consumes them is misspecified. When that map is of unknown fidelity—and the standard route

does not improve it—better inputs surface its bias rather than help. Both sets are scoped by the reference database that defines them, and neither is the deployment regime (composition audit, §S9), so the fidelity null is a null on small-molecule chemistry and the lever remains untested on drug-like solids. The useful form of the finding is one-sided and narrow: a place where grounding effort is *not* the remedy, on one row set of the map, rather than an allocation across chemistries—and an average over a set says only what that set happened to contain.

The unconstrained control’s representation. If the physics arm’s intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is not. A ridge probe of the control’s pair representation recovers the model’s own prediction at $R^2 = 0.98$ but not the measured ideal-solubility term—an upper bound at $R^2 \lesssim 0.30$ rather than an exclusion (limitation (ix)). What the representation is organised by instead is scaffold identity, consistent with the transfer limitation of §2.1.2: a test molecule on an unseen scaffold is not a small extrapolation but a point off the axis the representation is organised along. The seed-by-seed scores, the permutation nulls, the within-scaffold measure and its weakness, and the temperature probe are in §S10.

The contribution. The unit of contribution is the measurement, not any single number it returned. Its two ingredients are individually standard—the closure–insufficiency split is a law-of-total-variance identity, and the sign of grounding’s effect follows the model-discrepancy principle^{19,20}—and what is new is their combination into a usable instrument: an identity on a *fixed* operator, with the insufficiency side bounded rather than point-identified (§2.1.1), whose sign rule can be tested *a priori* and *bidirectionally*. The pKa domain supplies that test as an in-distribution sign flip (§3.5), at the resolution four series allow and no better—an oracle-on-a-fixed-operator affords the concept-leakage setting lacks, its in-

termediate having no ground truth.^{17,18} The solubility results are the instrument’s first application, and their transfer out of the matched regime is not established.

The accuracy confound, and what survives it. The physics and control arms were tuned separately—the settings differ by up to an order of magnitude and are listed in §S1—and no hyperparameter-matched control was run. A global ΔMAE of +0.14 between pipelines whose learning rates differ by an order of magnitude carries no information about the thermodynamic bottleneck, and neither do the per-solvent-class map, the per-solute-class map, the novelty inversion or the data-efficiency trend, all of which resolve the *same* difference along chemical axes. We therefore claim no accuracy result in either direction; those strata are reported in §S6.4 and §S6.5 as hypothesis-generating descriptions of where two particular pipelines differ, each carrying that caveat, and no conclusion rests on them. Running the matched control is the direct repair and we have not run it. What survives the confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning: §3.1, §3.2 (which uses no trained model at all), §3.3 and §3.4. Separately, and not as an accuracy claim about the bottleneck: we did not find a configuration in which the physics path attains a lower error than the control. External state-of-the-art models (FastSolv, SolProp) do no better on a by-solute split, which bounds the task’s difficulty and does not rank the models against one another (§S4), and grounding the crystal factor on external labels did not move the ceiling (§S6.3). The one place the physical *structure* helps is temperature extrapolation, as a hypothesis-class property the trained models do not exploit (§S6.1).

The sign rule’s own prediction. The penalty rule (Proposition S1) predicts *when* to expect these negatives: a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime. The synthetic sweep (Fig. S8; §S11) is true by construction there, so it verifies the instrument rather than the

hypothesis, while the pKa closure (§3.5) exercises the sufficiency channel on real data. Whether grounding helps is shaped by *two* axes, closure fidelity and how the latent is trained; characterising that plane on real systems is future work. The chemical strata form a pattern in the direction the sign rule would give; separately tuned pipelines can produce it for reasons unrelated to the prior and there is no multiplicity control, so §S6.4 records it as the experiment a matched control should run, not as evidence, and the diagnostic is *proposed* rather than validated.

Limitations: what the sets can carry.

(i) The phenomenon and its causal attribution lie in *different regimes*: the paradox is measured where the model operates (finite composition, all γ , $n=5608$), the attribution and the latent-departure result only on infinite-dilution activity data ($n=477$ across temperatures, and the 298 K, low- γ , VT-2005-matched $n=60$ and $n=44$ sets). Transfer back to the paradox’s own regime, argued from the in-regime signs, is a conjecture and is the principal open question here. The $n=60$ set is narrowly sourced—all by gas chromatography and all from three publications, 33 pairs from one—so neither a method-stratified nor a source-clustered re-estimate is available on it, and its margin is extraction-sensitive: a broader ThermoML pull at 298 K widens it but *inverts* it on the gas-chromatography subset, a sensitivity we leave unresolved and do not read as a verdict on the extraction. The broad set does not move the same way under the same cut (§3.2.1) and is not the set the paradox’s own oracle is defined on; the arithmetic, the enlarged-pull values and the matching-rule alternatives are in §S9. (ii) Under the deployed convention no *aggregate* margin is resolved on either set and only the literature-standard convention resolves one, while the headline convention is the deployed one (Table 2); the aggregate falls for the stronger reason of §3.2.1 rather than for want of resolution. (iii) Of the map’s six admissible cells the glycol ethers are the only row set also positive and not contained in another, and not the only cell whose deletions all run; the halo-

generated solvents and the mono-alcohols keep their sign under the one deletion each admits, so they are unverified rather than shown to fail.

Limitations: what the estimator can deliver.

(iv) B_{insuff} is upper-bounded, not point-identified (no replicates at fixed z^*); the smoothness-free Jensen bound assumes m and g share the $\ln \gamma_2^\infty$ reference state, established by the closure validation of §2.3, so the mean offset is decoder bias and not a units artifact. (v) The constant-offset margin is convention-specific and inverts under the deployed residual-only default on both sets; there the separation rests on the law-of-total-variance bound and the more-physics-worse effect. That bound is not itself convention-independent—only the estimand B_{insuff} is (§2.1.1)—so the deployed claim rests on it *computed under the deployed convention*. (vi) The stratification of §2.5 reads structure alone, so no cell was selected on its outcome, and it is exhaustive and disjoint *within* an axis but not across the two axes or their granularities: 362 of the 1711 stratum pairs share rows, all 37 water-as-solute rows lie inside three solvent classes, and 108 of the 129 alkane-solute rows are glycol-ether rows, so cells from different axes are not independent evidence and those rows are counted once (Table 4). Granularity is an analyst choice, which is why three solvent levels and two solute levels are all reported. There is no multiplicity control on the per-cell bootstrap frequencies, so one cell’s P describes that cell alone; the admissibility rule is the only protection in place, six of the nine solvent classes are not boundable here and three carry no estimate at all. The rule is one-sided in what it can deliver: because B_{insuff} is bounded only from above, no cell can ever return “better inputs would help here”, so the map only ever withholds a chemistry from an allocation of grounding effort.

Limitations: what is untested, and what is not auditable.

(vii) Absolute scaffold accuracy is encoder-bound but well *above* the inter-lab noise floor (0.15–0.31 in $\ln x_2$; §S6.1), so label noise does not explain the ceiling. (viii) The latent-departure result is three-seed

mean $\pm$ sd ($n=44$; §3.3): what it establishes is the *sign* and a transferable shared direction, the magnitudes ($51 \pm 2\%/72 \pm 1\%/4.1 \pm 0.5\times$) being reported but not stable across analysis configurations. The two top-two shares in play are distinct quantities: the learned-versus-reference deviation clears a smoothness-matched phase-randomised null but not a wrong-molecule one, and is measured on one grounded checkpoint rather than three seeds, whereas the isolation drift has no matched null of its own. (ix) The probe result on the unconstrained control is bounded by the same reference scarcity: its crystal-term and temperature tests rest on 13 held-out solutes and cannot be enlarged, so the negative is an upper bound at $R^2 \approx 0.30$ and not a demonstration that no crystal information is present; the crystal-slope test is additionally limited by the corpus, whose empirical slope carries $-\Delta H_{\text{fus}}/R$ at $r = +0.24$. The scaffold-organisation result needs no external reference, but its two partitions nearly coincide by construction, so a ratio near one is weak evidence of within-scaffold indistinguishability (§S10). (x) The closure-fidelity contrast compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself is not measured under a modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested; both need a typed (donor/acceptor) σ -profile head trained end to end through the 2010 kernel, and the solubility test a $\text{UD} \cap \text{BigSolDB}$ matched set. (xi) No difference in accuracy between the physics and control arms is attributable to the thermodynamic bottleneck; the confound and what survives it are stated above. (xii) The scaffold-leak guard is a property of the released stream builder and not a certified property of these runs: one σ -stream build on disk carries 91 held-out solutes, and per-run stream provenance was not logged (§S9). The paradox contrast is one checkpoint evaluated two ways and so cannot be generated by such a leak, and DirectGNN uses no stream, but auditability is not restored by those arguments. (xiii) Several of the numbers the verdict rests on are checkable only by re-running the released code, whose input artifacts are larger than the repository carries (§S9 names

which); the two decomposition sets are therefore deposited as row-level tables (Data Sets S1 and S2), so the sets and their margins can be recomputed without code, which does not deliver the provenance of the closure evaluations in them or of the trained arms.

4 Conclusions

Grounding a learned physical intermediate in reference values is two operations, not one, and only one of them helped here. Training the intermediate against reference values improved the model; supplying those values at prediction time did not. On the activity data where the two terms can be separated, they do not separate the same way in every chemistry: on one class of solvent, under a rule applied to every class alike, the bound on the fixed thermodynamic model’s own error lies above the bound on the contribution of its inputs, which is the condition under which more accurate inputs expose a model’s bias rather than remove it; on every other row set the separation is not established, at the sample sizes and the number of laboratories available. As a single number for a whole data set the comparison is an average over that set’s composition, a property of the set and not of the model. And the intermediate trained through that model stopped reporting the physical quantity it is named after, on the molecules where it can be held against its reference, which is the price of reading a grounded latent as a measurement.

The general statement is a design rule with a test attached. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures which of the two limits accuracy, and where it answers, the answer decides the remedy: a more faithful physical model, or better inputs. The answer comes chemistry by chemistry rather than once for a field, and it comes one-sided, so what the measurement returns is the chemistry on which better inputs are not the remedy—here one class of solvent, on which the standard route to a more faithful model did not clear the bar either. Whether a

latent trained through an imperfect model becomes, in general, a surrogate for that model’s error is the prediction this work leaves testable.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The analysis code, and those intermediate artifacts small enough to version, are available at <https://github.com/doctawho42/tgnn-solv> (commit [hash to be supplied at submission]) under the MIT licence, that commit being the one the reported numbers were produced from; the model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts—among them the two audit files §S9 names—will be deposited in a Zenodo archive, [whose DOI is registered at submission]. The data sources are public: BigSolDB 2.0²¹ for solubility, the VT-2005 database for the reference σ -profiles, Ref.²⁶ as redistributed with Ref.²³ for the UD profiles, ThermoML for the infinite-dilution activity coefficients, and the OPERA compilation²⁹ for the second-domain pKa data. Most analysis scripts carry their run commands in their docstrings, and the compute ledger (§S9) names the script and artifact file behind each audit. Four deposited artifacts carry disclosed defects, recorded in §S9 and §S3: the seed-44 oracle per-row file is truncated, the three-seed surrogate run’s per-run split provenance was not logged, one σ -stream build on disk was written against the wrong split files, and the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate.

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Supporting Information Available

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, ..., and every reference above that carries an S is to it. It contains supplementary methods; proofs; per-arm tables and decomposition robustness; the map’s remaining 44 strata; external baselines and ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa analysis; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. Two machine-readable tables are deposited with it: the 477-row broad IDAC set (Data Set S1, `broad_idac_set_477.csv`) and the 60-row VT-2005-matched set (Data Set S2, `vt2005_matched_set_60.csv`); their columns are listed in §S9.

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TOC Graphic

